สมาชิก

กัญญาวี ศรีเหรา 673380026-6 รับผิดชอบ Part 1

รสริน เมืองหงษ์ 673380289-4 รับผิดชอบ Part 2

สโรชา เสาทอง 673380296-7 รับผิดชอบ Part 3

ปวิศร์ แพงมา 673380047-8 รับผิดชอบ Part 5

ธนกร ทองศรี 673380040-2 รับผิดชอบ Part 4 และ Worksheet

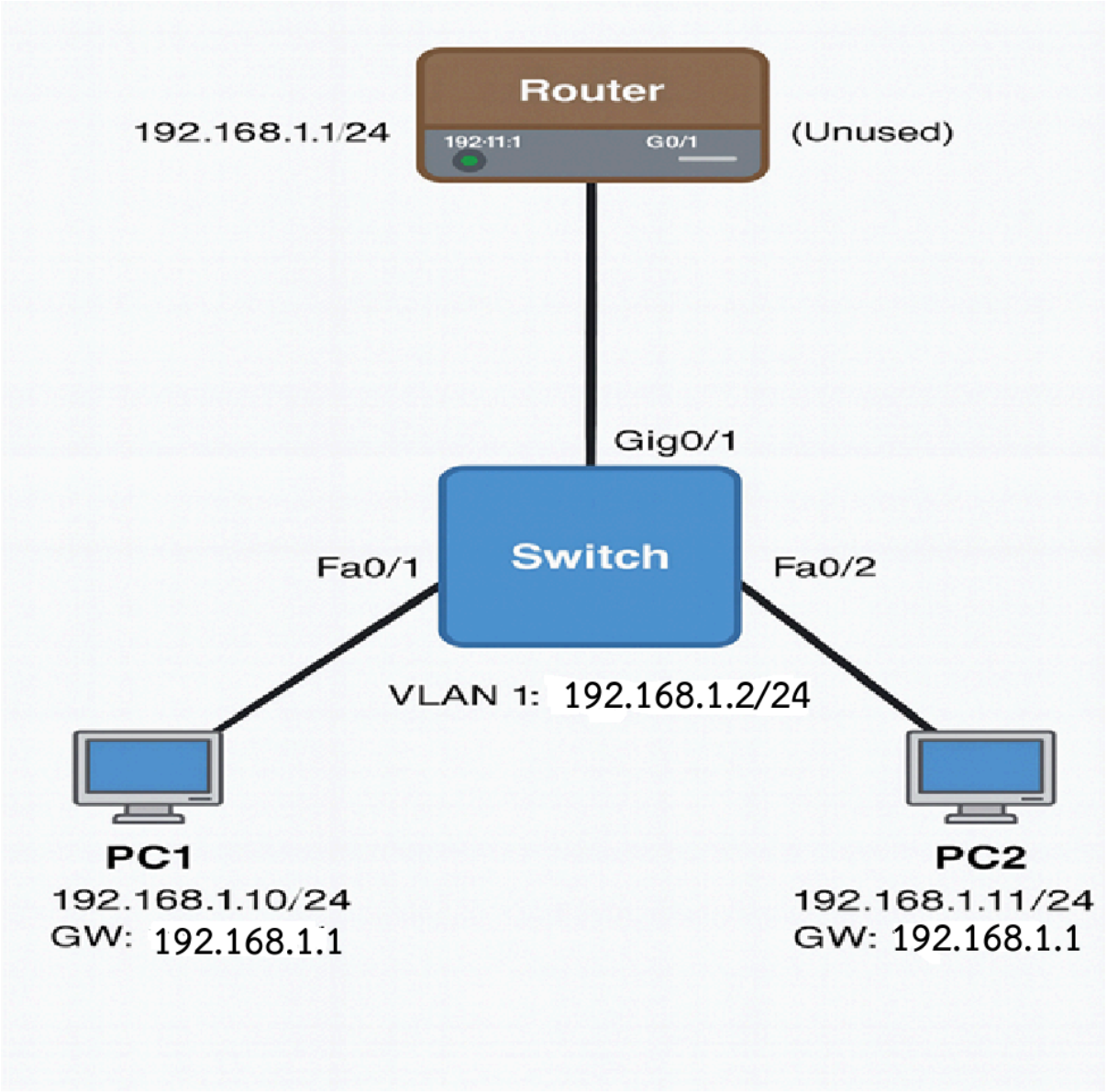
ไฟล์ :
https://drive.google.com/file/d/1s_MYfcr02kry2ZZw0sxl2G5hdwNrzoEz/view?usp=drive_link

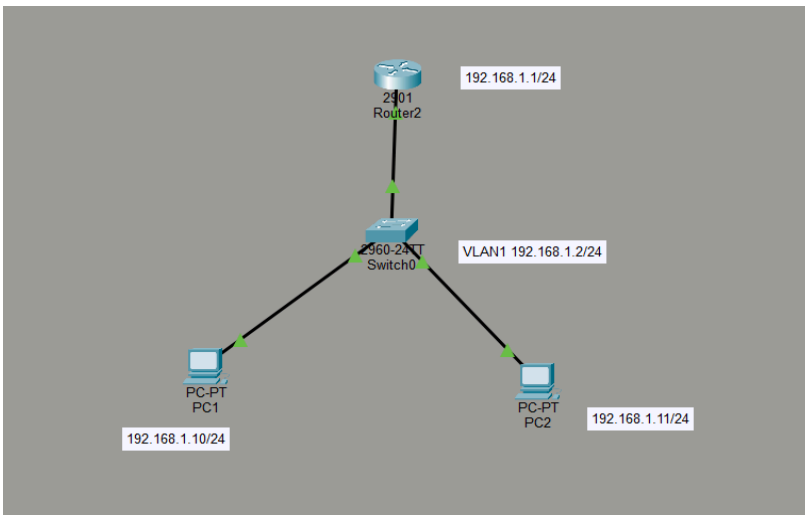
Lab 1: Complete Lab-Focused Teaching Package

📊 TOPOLOGY DIAGRAM

None

```
                    192.168.1.1/24
            [G0/0]--[Router]--[G0/1] (Unused)
                        |
                        |
                 [Gig0/1] (Switch Port)
                        |
            [Switch] (VLAN 1: 192.168.1.2/24)
                 /              \
                /                \
        [Fa0/1]                [Fa0/2]
            |                      |
            |                      |
      [PC1]                    [PC2]
192.168.1.10/24         192.168.1.11/24
GW: 192.168.1.1        GW: 192.168.1.1
```

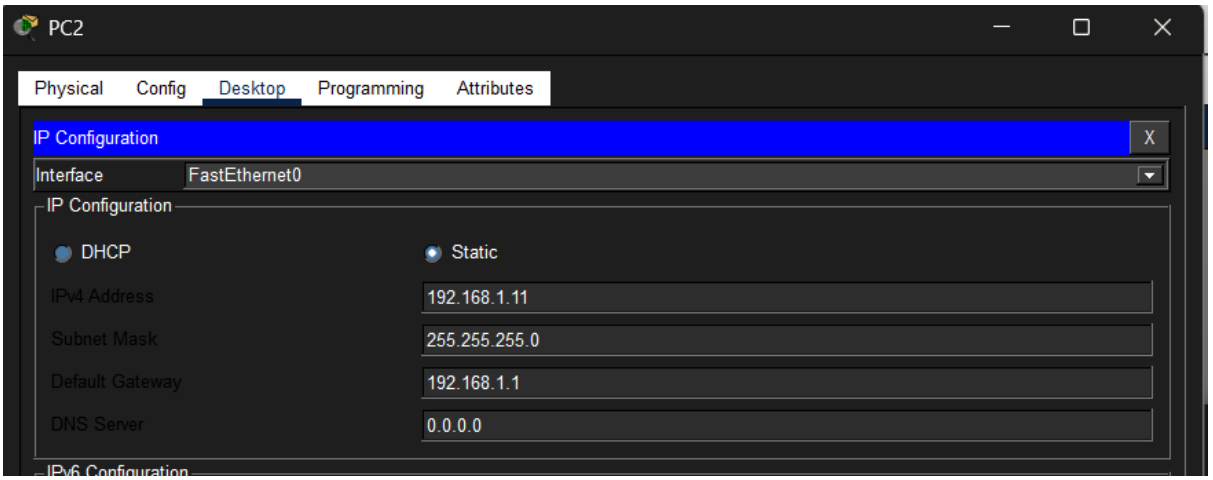

Router
192.168.1.1/24
192.11.1
G0/1
(Unused)
Gig0/1
Switch
Fa0/1
Fa0/2
VLAN 1: 192.168.1.2/24
PC1
192.168.1.10/24
GW: 192.168.1.1
PC2
192.168.1.11/24
GW: 192.168.1.1

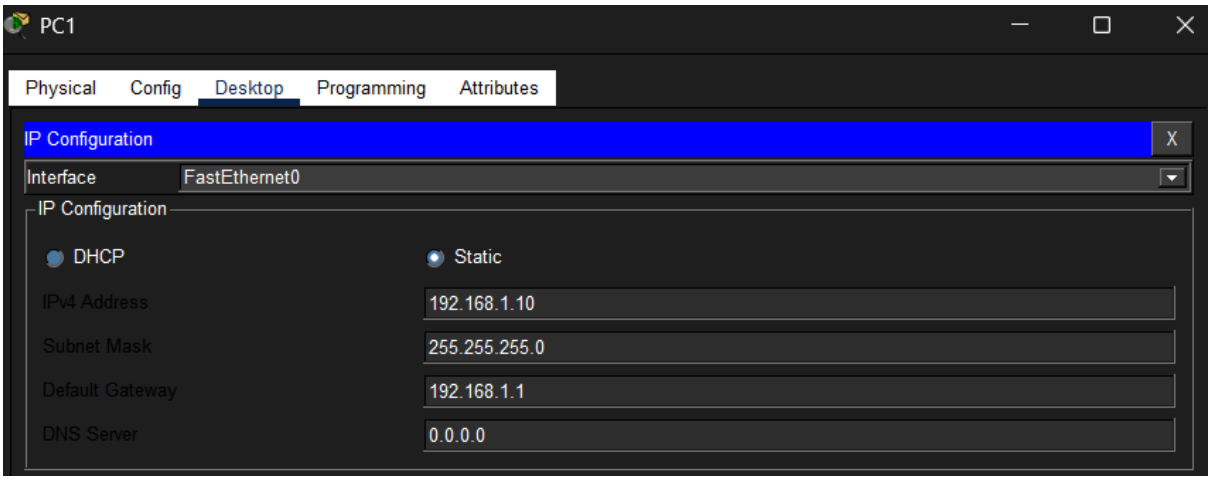

2901
Router2
192.168.1.1/24
2960-24TT
Switch0
VLAN1 192.168.1.2/24
PC-PT
PC1
192.168.1.10/24
PC-PT
PC2
192.168.1.11/24

📋 ADDRESSING TABLE

Device	Interface	IP Address	Subnet Mask	Default Gateway	MAC (Dynamic)
R1	G0/0	192.168.1.1	255.255.255.0	N/A	0001.9643.AC01
S1	VLAN 1	192.168.1.2	255.255.255.0	192.168.1.1	0002.1745.8B02
PC1	NIC	192.168.1.10	255.255.255.0	192.168.1.1	0050.7966.6803
PC2	NIC	192.168.1.11	255.255.255.0	192.168.1.1	0090.2B14.7C04

Network: 192.168.1.0/24
Usable Host Range: 192.168.1.1 - 192.168.1.254
Broadcast: 192.168.1.255





🎯 LAB OBJECTIVES

- **Part 1:** Configure basic device addressing (R1, S1, PC1, PC2)
- **Part 2:** Verify connectivity and observe encapsulation
- **Part 3:** Analyze protocol headers using simulation
- **Part 4:** Troubleshoot layered connectivity issues
- **Part 5:** Document protocol interactions

🔧 LAB INSTRUCTION STEPS

PART 1: DEVICE CONFIGURATION (30 minutes)

Step 1: Router Configuration

```
None
Router> enable
Router# configure terminal
Router(config)# hostname R1
R1(config)# interface gigabitEthernet 0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# description LINK_TO_SWITCH_S1
R1(config-if)# exit
R1(config)# ip route 0.0.0.0 0.0.0.0 null0  ! Static default route for
completeness
R1(config)# end
R1# copy running-config startup-config
```

```
Router>enable
Router# configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#interface gigabitEthernet 0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to
up

R1(config-if)#description LINK_TO_SWITCH_S1
R1(config-if)#exit
R1(config)#ip route 0.0.0.0 0.0.0.0 null0
%Default route without gateway, if not a point-to-point interface, may impact
performance
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
```

Step 2: Switch Configuration

None

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname S1
S1(config)# interface vlan 1
S1(config-if)# ip address 192.168.1.2 255.255.255.0
S1(config-if)# no shutdown
S1(config-if)# exit
S1(config)# ip default-gateway 192.168.1.1
S1(config)# interface range fastEthernet 0/1-2
S1(config-if-range)# switchport mode access
S1(config-if-range)# switchport access vlan 1
S1(config-if-range)# no shutdown
S1(config-if-range)# exit
S1(config)# end
S1# copy running-config startup-config
```

Switch0

Physical Config CLI Attributes

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up


Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname S1
S1(config)#interface  vlan 1
S1(config-if)#ip address 192.168.1.2 255.255.255.0
S1(config-if)#no shutdown

S1(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

S1(config-if)#ip default_gateway 192.168.1.1
                 ^
% Invalid input detected at '^' marker.

S1(config-if)#ip default-gateway 192.168.1.1
S1(config)#interface range fastEthernet 0/1-2
S1(config-if-range)#switchport mode access
S1(config-if-range)#switchport access vlan1
                                         ^
% Invalid input detected at '^' marker.

S1(config-if-range)#switchport access vala 1
                                     ^
% Invalid input detected at '^' marker.

S1(config-if-range)#switchport access vlan 1
S1(config-if-range)#no shutdown
S1(config-if-range)#exit
S1(config)#end
S1#
%SYS-5-CONFIG_I: Configured from console by console

S1#copy run
% Incomplete command.
S1#copy running-config start
Destination filename [startup-config]?
Building configuration...
[OK]
S1#
```

Step 3: PC Configuration

PC1:

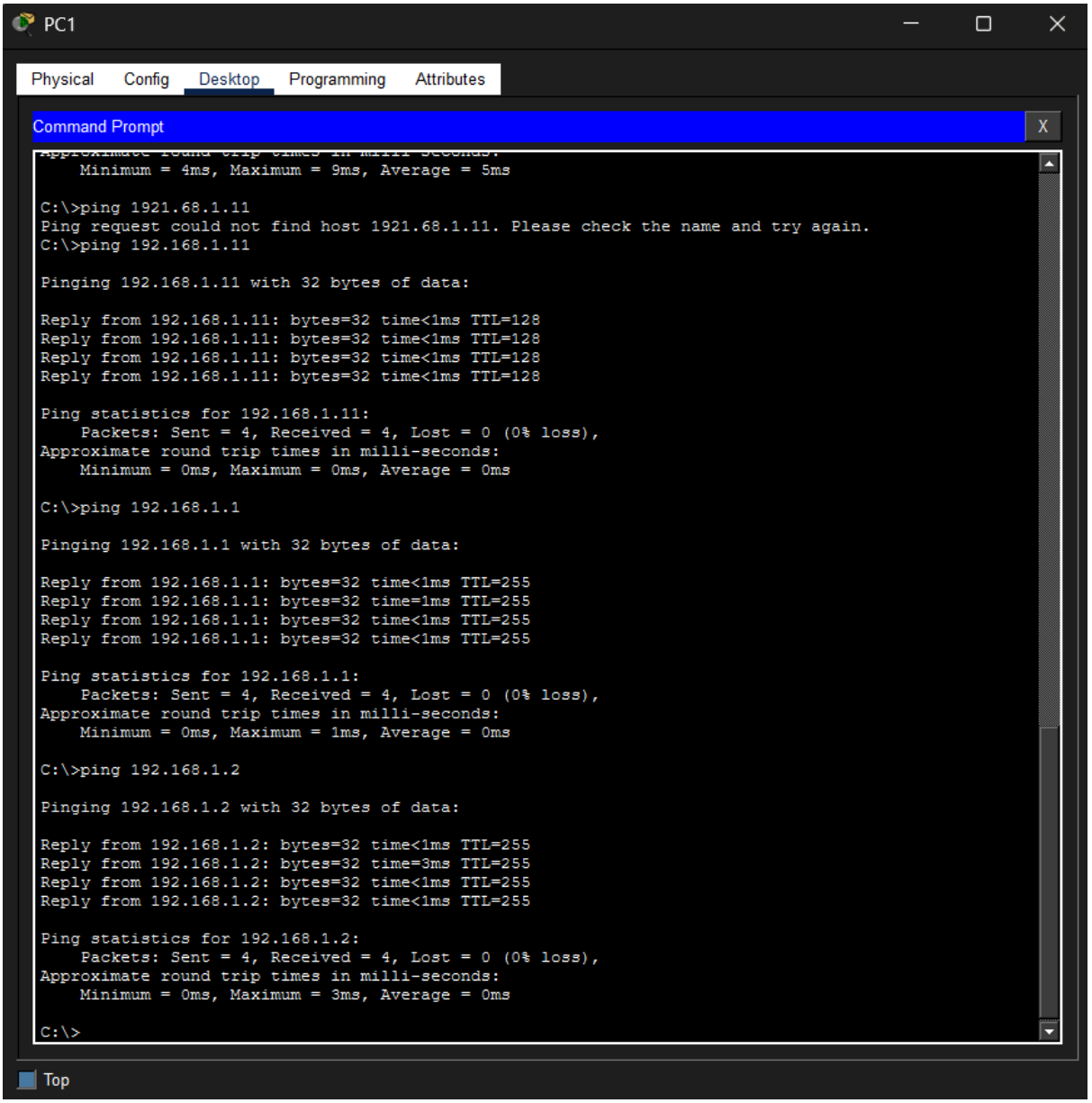
- IP Address: 192.168.1.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1

PC2:

- IP Address: 192.168.1.11
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1

PART 2: CONNECTIVITY VERIFICATION (20 minutes)

Step 4: Basic Connectivity Tests



None

```
! From PC1 Command Prompt:
PC1> ping 192.168.1.11  ! Verify PC1 to PC2 connectivity
PC1> ping 192.168.1.1   ! Verify PC1 to R1 connectivity
PC1> ping 192.168.1.2   ! Verify PC1 to S1 connectivity

! From R1:
R1# show ip interface brief
! Verify G0/0 status: Protocol UP, Line Protocol UP
! IP address correctly assigned

! From S1:
S1# show vlan brief
! Verify VLAN 1 exists and ports Fa0/1-2 are assigned
```

```
S1# show mac address-table
! Observe MAC addresses learned on Fa0/1 and Fa0/2
```

```
R1>
R1>enable
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1#
```

```
S1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
S1#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0004.9a62.6e66    DYNAMIC     Fa0/1
   1    0060.3e53.7a70    DYNAMIC     Fa0/2
   1    00d0.ffd8.7a01    DYNAMIC     Gig0/1
S1#
```

Step 5: ARP Table Examination

```
None
! From PC1 Command Prompt:
PC1> arp -a
! Record the ARP table entries
! Expected: 192.168.1.1 and 192.168.1.11 should appear with MAC addresses

! From R1:
R1# show arp
! Verify PC1 and PC2 MAC addresses are in ARP table
```

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.1           00d0.ffd8.7a01        dynamic
  192.168.1.11          0060.3e53.7a70        dynamic
```

```
R1>enable
R1#show arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  192.168.1.1            -    00D0.FFD8.7A01  ARPA   GigabitEthernet0/0
Internet  192.168.1.10          13    0004.9A62.6E66  ARPA   GigabitEthernet0/0
R1#
```

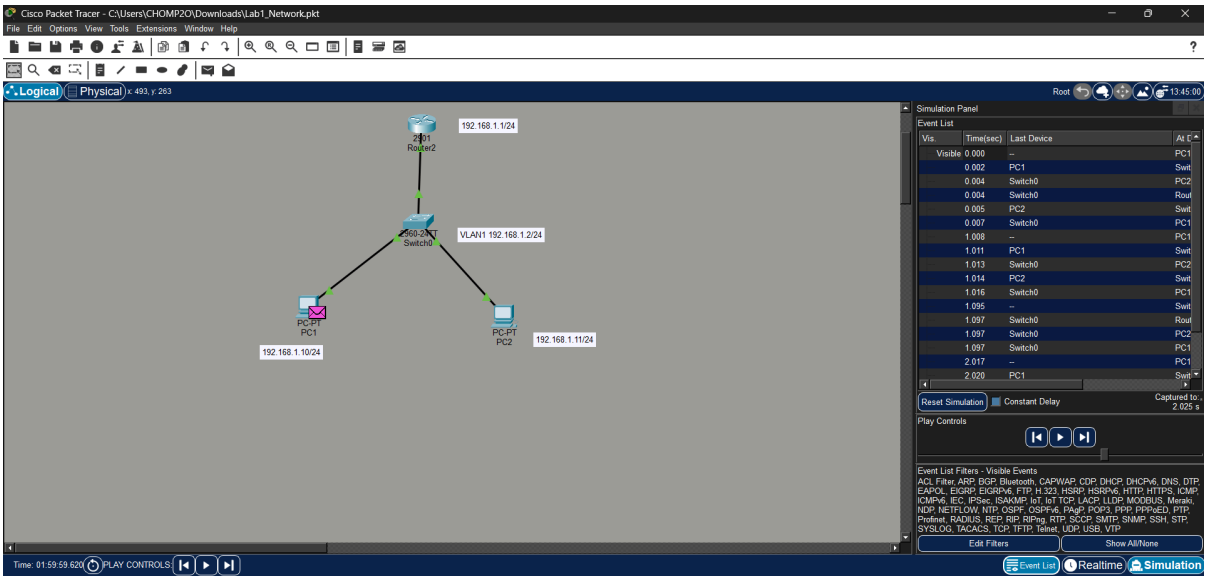
PART 3: ENCAPSULATION ANALYSIS (25 minutes)

Step 6: Packet Tracer Simulation Mode

1. Enter **Simulation Mode** in Packet Tracer
2. Create a **PDU** from PC1 to PC2 (ICMP Echo)
3. **Slow the simulation** and observe each step:

Observation Points:

- **Layer 2:** Ethernet frame construction (Source/Dest MAC)
- **Layer 3:** IP header creation (Source/Dest IP)
- **ARP Process:** How PC1 resolves PC2's MAC address
- **Switch Forwarding:** How S1 uses MAC table
- **Encapsulation/Decapsulation** at each hop



Step 7: Protocol Header Documentation

Create this table in your lab report:

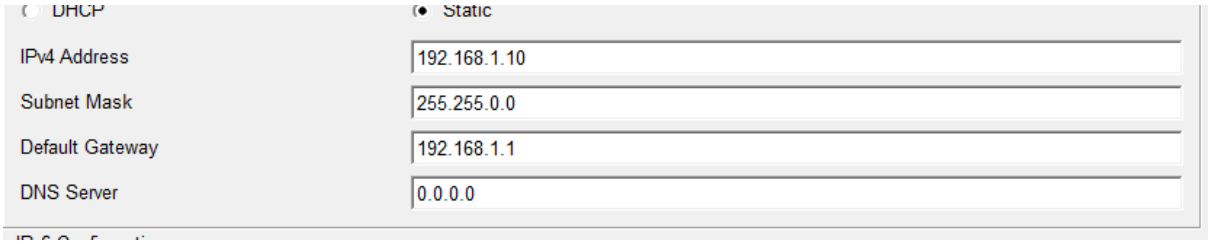
Step	Protocol	OSI Layer	TCP/IP Layer	Key Field Observed
1	ARP Request	Layer 2	Network Access	Target IP: 192.168.1.11
2	ARP Reply	Layer 2	Network Access	Sender MAC: [PC2's MAC]
3	ICMP Echo	Layer 3	Internet	Type: 8, Code: 0
4	Ethernet	Layer 2	Network Access	Frame Check Sequence
5	ICMP Reply	Layer 3	Internet	Type: 0, Code: 0

PART 4: TROUBLESHOOTING CHALLENGE (25 minutes)

Step 8: Intentional Break-Fix Scenarios

Scenario A: Incorrect Subnet Mask

```
None
! On PC1, change to wrong subnet mask:
PC1> ipconfig /ip 192.168.1.10 255.255.0.0 192.168.1.1
! Attempt ping to PC2 - Observe failure
! Use 'arp -a' to diagnose
! Fix: Restore correct mask 255.255.255.0
```



```
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

แม้จะมีการเปลี่ยนค่า Subnet Mask แต่การ Ping ระหว่างอุปกรณ์ยังสามารถทำได้ เนื่องจากSubnet Mask ที่เลือกใช้ยังครอบคลุมช่วง IP ของอุปกรณ์ทั้งสองอยู่ การ Ping จึงยังคงสามารถทำงานได้ตามปกติ

Scenario B: Router Interface Shutdown

```
None
! On R1:
R1# configure terminal
R1(config)# interface g0/0
R1(config-if)# shutdown
! From PC1: ping 192.168.1.1 - Observe failure
! Use 'tracert' or examine ARP table
! Fix: R1(config-if)# no shutdown
```

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface g0/0
R1(config-if)#shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to down
```

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>tracert
Cisco Packet Tracer PC Tracert

Usage: tracert target

C:\>
```

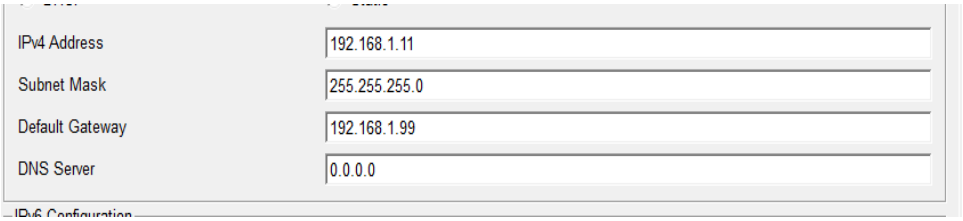
```
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
```

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

เมื่อสั่ง shutdown Router จะทำให้พอร์ตถูกปิดและไม่สามารถสื่อสารกับเครือข่ายได้ ส่งผลให้ PC1 ไม่สามารถ ping ได้ จนกว่าจะเปิดพอร์ตอีกครั้งด้วยคำสั่ง no shutdown

Scenario C: Wrong Default Gateway

```
None
! On PC2:
PC2> ipconfig /dg 192.168.1.99
! Attempt ping from PC2 to R1
! Use 'route print' (Windows) to verify gateway
! Fix: Restore to 192.168.1.1
```




```
C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time=1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

แม้จะเปลี่ยนค่า Default Gateway แต่การ Ping ระหว่าง PC2 และ PC1 ยังสามารถทำได้ เนื่องจากทั้งสองอุปกรณ์อยู่ในเครือข่ายเดียวกัน การสื่อสารจึงเกิดขึ้นโดยตรงผ่าน Switch โดยใช้ ARP โดยไม่ต้องผ่าน Default Gateway ซึ่งจะถูกใช้งานเฉพาะเมื่อมีการติดต่อไปยังเครือข่ายอื่นเท่านั้น

PART 5: ADVANCED OBSERVATION (15 minutes)

Step 9: Protocol Stack Verification

```
None
! From PC1, generate extended ping with different protocols:
PC1> ping 192.168.1.11 -n 5 -l 1500  ! Test MTU/fragmentation
PC1> arp -d *  ! Clear ARP cache, then ping to observe ARP process

! On S1, observe MAC table changes:
S1# clear mac address-table dynamic
S1# show mac address-table  ! Observe empty table
! Generate traffic, then check again
```

เนื่องจาก ping 192.168.1.11 -n 5 -l 1500 ไม่สามารถใช้ได้ ทำการใช้ ping 192.168.1.11 แทน

```
C:\>ping 192.168.1.11 -n 5 -l 1500
Invalid Command.

C:\> ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time=3ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 0ms
```

เคลียร์ ARP Table

```
C:\>arp -d
C:\>arp -a
No ARP Entries Found
```

แล้ว ping ใหม่อีกครั้ง

```
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time=7ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time=1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 7ms, Average = 2ms

C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.11          0001.427d.e4a7        dynamic
```

ทำการล้าง MAC table

```
S1#clear mac address-table dynamic
S1#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----
```

แล้วทำการเช็ค MAC table อีกครั้งหลัง ping

```
S1#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0001.427d.e4a7    DYNAMIC     Fa0/2
   1    0001.c92c.3c01    DYNAMIC     Gig0/1
   1    0009.7c57.6c11    DYNAMIC     Fa0/1
S1#
```

Step 10: Documentation

เมื่อผู้ใช้สั่งคำสั่ง ping ระบบจะสร้าง ICMP Echo Request จากนั้นนำข้อมูล ICMP ใส่ลงใน IP Packet ต่อมา IP Packet จะถูกห่อหุ้มเป็น Ethernet Frame โดยใช้ MAC Address ที่ได้จากกระบวนการ ARP และสุดท้ายข้อมูลจะถูกส่งออกไปยังเครือข่ายผ่านสื่อทางกายภาพ เช่น สาย LAN

📝 LAB DELIVERABLES

Worksheet 1: Configuration Verification

Checkpoint	Command Used	Expected Result	Actual Result	Pass/Fail
R1 Interface Status	`show ip int brief`	G0/0 UP/UP	GigabitEthernet0/0 192.168.1.1 YES manual up up	Pass
S1 VLAN Config	`show vlan brief`	VLAN 1 Active	1 default active Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2	Pass
PC1 Connectivity	`ping 192.168.1.11`	Success (4 replies)	Ping statistics for 192.168.1.11: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)	Pass
ARP Resolution	`arp -a` on PC1	2+ entries	Internet Address Physical Address Type 192.168.1.1 0002.1751.a601 dynamic 192.168.1.2 0060.709d.a305 dynamic	Pass

Checkpoint	Command Used	Expected Result	Actual Result	Pass/Fail
			192.168.1.11 0060.5c03.55ea dynamic	
MAC Learning	`show mac address-table` on S1	2 dynamic entries	Vlan Mac Address Type Ports ---- ----------- -------- ----- 1 0002.1751.a601 DYNAMIC Gig0/1 1 0060.3e4e.ac93 DYNAMIC Fa0/1 1 0060.5c03.55ea DYNAMIC Fa0/2	Pass

Worksheet 2: Protocol Analysis

Capture this data during simulation:

1. **ARP Request Frame:**

    - Source MAC: MAC :0009.7C57.6C11
    - Destination MAC: FF:FF:FF:FF:FF:FF
    - ARP Opcode: 1

2. **ICMP Echo Packet:**

    - Time to Live (TTL): 128
    - Protocol Field: ICMP (Value = 1)

3. **Encapsulation Chain:** Data → ICMP Header → IP Header → Ethernet Header → Bits

Worksheet 3: Troubleshooting Journal

Problem Injected	Symptoms Observed	Diagnostic Commands Used	Root Cause	Solution
Wrong Subnet Mask	ยังสามารถ ping ได้	`ping`, `arp -a`	ตั้งค่า Subnet Mask ไม่ถูกต้อง ทำให้อุปกรณ์ถูก	แก้ไข Subnet Mask ให้ถูกต้อง

Problem Injected	Symptoms Observed	Diagnostic Commands Used	Root Cause	Solution
			จัดอยู่คนละเครือข่าย	
Interface Shutdown	ไม่สามารถ ping ได้	`show ip int brief`	Router อยู่ในสถานะ shutdown	เปิดใช้งาน interface ด้วยคำสั่ง no `shutdown`
Incorrect Gateway	ยังสามารถ ping อุปกรณ์อยู่ในเครือข่ายเดียวกัน	`ipconfig`, `tracert`	ตั้งค่า Default Gateway ไม่ถูกต้อง	ตั้ง Default Gateway ให้ตรงกับ IP ของ Router

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🎓 ASSESSMENT QUESTIONS

1. When PC1 pings PC2, which protocol resolves the MAC address before ICMP operates?
   ANS โปรโตคอล ARP

2. What would happen if you changed the subnet mask on PC1 to 255.255.255.128?
   ANS การ ping จะไม่สำเร็จ เนื่องจากอยู่คนละ subnet

3. Why does the switch not need an IP address for basic frame forwarding?
   ANS Switch ใช้ MAC Address ในการทำงาน ไม่ใช้ IP Address

4. Which OSI layers are involved when R1 forwards a packet between interfaces?
   ANS Layer 3 และ และ Layer 2

5. What is the purpose of the TTL field in the IP header you observed?
   ANS กำหนดอายุการใช้งานของแพ็กเก็ตในเครือข่าย เพื่อป้องกันไม่ให้แพ็กเก็ตวนลูปอยู่ในเครือข่ายตลอดไป